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A note on the geometric series as a species frequency model

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SUMMARY

We consider a certain randomized geometric series of relative abundances and the limit of the Dirichlet distribution used in the derivation of the logarithmic series distribution. When an individual is chosen at random from either of the populations, the abundance of the species it represents is shown to have the same distribution. The maximum likelihood estimate for the geometric series model with fixed abundances is evaluated, and a bias correction is proposed.

Some key words: Abundance of species; Ecology; Geometric series; Logarithmic series distribution; Maximum likelihood.

1. A RELATION TO FISHER'S LOGARITHMIC SERIES

The well-known logarithmic series distribution derived by Fisher, Corbet & Williams (1943) has proved useful as a species frequency distribution. It is obtained by assuming that the abundances of the different species, say $x_1, \ldots, x_s$, are realizations of independent gamma distributions

$$\frac{\rho^k}{\Gamma(k)} x^{k-1} e^{-\rho x} \quad (k, \rho > 0). \tag{1}$$

Fisher considered the limit $k \to 0$, $s \to \infty$, and such that ks approaches a constant α. Though the abundances of each particular species tend to zero in probability, the distribution of $x_1 + \ldots + x_s$ tends to $\rho^\alpha x^{\alpha-1} e^{-\rho x}/\Gamma(\alpha)$. The abundance of the whole population is therefore positive. This is related to a theorem of Kingman (1975) implying that the order statistics of the relative abundances, $p_i = x_i/\Sigma x_j$, in the above case possess a limiting distribution.

If k is positive, the relative abundances have the Dirichlet distribution

$$\frac{\Gamma(ks)}{\{\Gamma(k)\}^s} (p_1 \ldots p_s)^{k-1} \quad (\Sigma p_i = 1). \tag{2}$$

Now let an individual be chosen at random from the population and let X denote the abundance of the species it represents. In general we will say that two sequences of random relative abundances are equal in distribution if the corresponding distributions of X are identical. For the above Dirichlet distribution we find

$$E(X^r) = E\{E(X^r | p_1, \ldots, p_s)\} = E(\Sigma p_i^{r+1}) = \frac{\Gamma(ks+1)\,\Gamma(k+r+1)}{\Gamma(k+1)\,\Gamma(ks+r+1)},$$

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so that, as $k \to 0$, $ks \to \alpha$,

$$\lim E(X^r) = \frac{\Gamma(\alpha+1)\,\Gamma(r+1)}{\Gamma(\alpha+r+1)} \quad (r = 0, 1, \ldots). \tag{3}$$

In practice it is most natural to consider the abundances as fixed, and several attempts have been made to define sequences of relative abundances that are like realizations of (2) (Holgate, 1969; Engen 1974; Watterson, 1974). A result of Watterson (1974) and of Odum, Cantlon & Kornicker (1960) indicates that such models of fixed relative abundances are likely to be approximately the geometric series

$$p_i = \theta(1-\theta)^{i-1} \quad (i = 1, 2, \ldots), \tag{4}$$

where θ is approximately α^{-1}. We now give a further demonstration of the relation between (4) and the limiting form of (2). Let $P_1, P_2, \ldots$ be a sequence of independent beta variates with the same distribution $\alpha(1-x)^{\alpha-1}$, and consider the sequence

$$Q_1 = P_1, \quad Q_i = P_i \prod_{j=1}^{i-1}(1-P_j) \quad (i = 2, 3, \ldots). \tag{5}$$

If $\theta = 1/(\alpha+1)$, then $E(Q_i) = \theta(1-\theta)^{i-1} = p_i$. Therefore the Q_i's may be considered as a sequence of random relative abundances whose expectations equal those specified in the geometric series model. Since $Q_1 + \ldots + Q_j \leqslant 1$ and $\lim E(Q_1 + \ldots + Q_j) = 1$, $Q_1 + \ldots + Q_j$ tends to one in probability, so that the Q_i's may denote relative abundances. Again considering the moment sequence corresponding to (3), we get

$$\sum_{i=1}^{\infty} E(Q_i^{r+1}) = \frac{\Gamma(\alpha+1)\Gamma(r+1)}{\Gamma(\alpha+r+1)},$$

which is equivalent to (3). Thus, we have established the following result: as $k \to 0$, $s \to \infty$, with $ks \to \alpha$, the sequence of random relative abundances given by the Dirichlet distribution (2) tends in distribution to that specified in (5). In the light of the success of Fisher's model, the geometric series therefore seems to be a very reasonable model of fixed relative abundances.

2. The maximum likelihood estimate

Let the species be recognizable and numbered as $S_1, S_2, \ldots$. Here the indices bear no relation to the ordering of the abundances. If q_i is the abundance of S_i our model is $q_i = \theta(1-\theta)^{r_i-1}$, where $R = (r_1, r_2, \ldots)$ is the set of all positive integers in some order. To be realistic we must assume that R is unknown. As a sample of size N is drawn from the population the likelihood function is

$$L = \sum_{i=1}^{\infty} X_i \log\{\theta(1-\theta)^{r_i-1}\},$$

where X_i is the number of individuals in the sample belonging to S_i, and R as well as θ are unknown parameters. Because the ordering of the p_i's is the same for all θ, $p_1 > p_2 > \ldots$, L is maximized for a given θ by putting r_i equal to the index number of the ith most abundant species in the sample. Thus, if the order statistics of the X_i's are $X^* = (X^{(1)}, X^{(2)}, \ldots)$ and $T = (t_1, t_2, \ldots)$ is defined by $X^{(1)} = X_{t_1} \geqslant X^{(2)} = X_{t_2} \geqslant \ldots$, the estimate of R is $\hat{R} = T$. Maximizing with respect to $\alpha = (1-\theta)/\theta$, we get the maximum likelihood estimate $\hat{\alpha} = (1/N)\Sigma i X^{(i)} - 1$. Define further $Y = (Y_1, Y_2, \ldots)$ by $Y_{r_i} = X_i$, so that $E(Y_i) = Np_i$. Then, if R was known, the maximum likelihood estimate would be $\tilde{\alpha} = N^{-1}\Sigma i Y_i - 1$, which is easily seen to be unbiased. Comparing $\hat{\alpha}$ with $\tilde{\alpha}$, we see that Y is replaced by X^* when R is unknown. Intuitively, this may cause a considerably negative bias in $\hat{\alpha}$, and it seems necessary to introduce a correction. It is shown below that the bias is of order N^{-1}, so that we can correct for the first term by Quenouille's jackknife method (Quenouille, 1956) by considering subsamples of size $N-1$. The corrected estimate takes the simple form

$$\hat{\alpha}' = \frac{1}{N} \sum_{i \leqslant j} i n_i n_j - 1,$$

where n_i is the number of species represented by i individuals in the sample. It is also easy to verify that $N \operatorname{var}(\tilde{\alpha}) = (1-\theta)/\theta^2$, and by arguments similar to those above one can show that

$$\lim N \operatorname{var}(\hat{\alpha}') = \lim N \operatorname{var}(\hat{\alpha}) = \lim N \operatorname{var}(\tilde{\alpha}).$$

Now consider the bias of $\hat{\alpha}$. Let $U^{(i)}$ be the order statistics of the Y_i's omitting Y_1, and write

$$E(\Sigma i X^{(i)}) = N(\alpha+1) + C(N, \theta),$$

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where $C(N,\theta)$ is a bias function. Write $\beta = \Sigma i(X^{(i+1)} - U^{(i)})$. Then $\Sigma i X^{(i)} = N + \Sigma i X^{(i+1)} = N + \Sigma i U^{(i)} + \beta$, and taking expectations we get $E\{C(N,\theta) - C(N-Y_1,\theta)\} = E(\beta)$. Now, there exists an integer r so that $E(Y_1) > E(Y_r + Y_{r+1} + \ldots) = E(W_r)$. Defining the events $A_i = (Y_1 > Y_i)$, $A_r = (Y_1 > W_r)$ $(i = 2, \ldots, r-1)$, we have that

$$\mathrm{pr}(Y_1 > Y_2, Y_3, \ldots) \geqslant \mathrm{pr}\left(\bigcap_{i=2}^{r} A_i\right) = \mathrm{pr}(A_r) \prod_{i=2}^{r-1} \mathrm{pr}\left(A_i \,\middle|\, \bigcap_{j=i+1}^{r} A_j\right) \geqslant \prod_{i=2}^{r} \mathrm{pr}(A_i).$$

As N tends to infinity, $Y_1, \ldots, Y_{r-1}, W_r$ are asymptotically normally distributed, and by the well-known asymptotic expression for the normal integral

$$\int_x^\infty \frac{1}{\sqrt{(2\pi)}} e^{-\frac{1}{2}x^2} dx = \frac{1}{\sqrt{(2\pi)}} e^{-\frac{1}{2}x^2} (1/x - 1/x^3 + \ldots)$$

we get $1 - \mathrm{pr}(Y_1 > Y_2, Y_3, \ldots) = o(N^{-\frac{1}{2}} e^{-\phi(\theta)N})$, where $\phi(\theta)$ is some positive function of θ. Since $\beta \leqslant N$ and $(Y_1 > Y_2, Y_3, \ldots)$ implies that $\beta = 0$, it follows that $E(\beta) = o(N^{\frac{1}{2}} e^{-\phi(\theta)N})$, or $\lim E(\beta) = 0$, as $N \to \infty$. Thus $\lim E[C(N,\theta) - C\{N(1 - Y_1/N), \theta\}] = 0$. Since Y_1/N tends to θ in probability, $C(N,\theta)$ is at least a periodic function of $\log N$ in the limit, which is sufficient to conclude that $C(N,\theta)$ is of order one. The bias of the maximum likelihood estimate is therefore of order $1/N$.

This conclusion could not be drawn immediately since our estimate is not a function of the sample k statistics that can be expanded in Taylor series.

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On the modes of a mixture of two von Mises distributions

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Summary

The number of modes of a mixture of two von Mises distributions is examined. A method of calculating the positions of the modes and antimodes is also given.

Some key words: Bimodal circular distributions; Bimodality; Detection of modes and antimodes; Directional data; Mixture; von Mises distribution.

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